

MAT 350

Engineering mathematics

Lecture -2

Variable Separable, Exact ODE with modellings

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First Order Ordinary Differential Equations:

Separable Equations:

A first-order differential equation of the form

$$\frac{dy}{dx} = g(x)h(y)$$

is said to be separable or to have separable variables, and the equation is called separable equation.

Examples:

$$(1) \quad \frac{dy}{1 + y^2} = dx.$$

$$(2) \quad (1 + x) dy - y dx = 0.$$

$$(3) \quad \frac{dy}{dx} = -\frac{x}{y}, \quad y(4) = -3.$$

First Order Ordinary Differential Equations:

Separable Equations:

Solution Technique: When a first order ODE is separable variables, the solution can be found by integration, such that

$$p(y)dy = g(x)dx, \quad \text{where } p(y)=1/h(y)$$

$$\int p(y)dy = \int g(x)dx + c$$

$$H(y) = G(x) + c,$$

First Order Ordinary Differential Equations:

$$\text{Solve } (1 + x) dy - y dx = 0.$$

SOLUTION: Dividing by $(1+x)y$, we can write

$$\frac{dy}{y} = \frac{dx}{1+x}$$

from which it follows that

$$\int \frac{dy}{y} = \int \frac{dx}{1+x}$$

$$\ln|y| = \ln|1+x| + c_1$$

$$y = e^{\ln|1+x|+c_1} = e^{\ln|1+x|} \cdot e^{c_1}$$

$$= |1+x| e^{c_1}$$

$$= \pm e^{c_1}(1+x).$$

Relabeling $\pm e^{c_1}$ as c then gives $y = c(1+x)$.



First Order Ordinary Differential Equations:

Separable Equations:

$$\text{Solve } y' = (x + 1)e^{-x}y^2$$

Solution: We obtain by side-changing

$$y^{-2} dy = (x + 1)e^{-x} dx.$$

This means the ODE is separable. Then by integration,

$$-y^{-1} = -(x + 2)e^{-x} + c,$$

$$y = \frac{1}{(x + 2)e^{-x} - c}.$$

If an initial condition $y(0)=1$ is introduced with the ODE, then we find the value of the constant c , which gives $c=1$.

$$y = \frac{1}{(x + 2)e^{-x} - 1}$$



Exercise 2.2, Zill

$$25. \quad x^2 \frac{dy}{dx} = y - xy, \quad y(-1) = -1$$

Solution: Hints

$$\frac{1}{y} dy = \frac{1-x}{x^2} dx = \left(\frac{1}{x^2} - \frac{1}{x} \right) dx$$

Integrating both sides,

$$\begin{aligned} \text{we obtain } \ln |y| &= -\frac{1}{x} - \ln |x| = c \\ \text{or } xy &= c_1 e^{-1/x}. \end{aligned}$$

Substituting the initial condition

$$y(-1) = -1 \text{ we find } c_1 = e^{-1}.$$

The solution of the initial-value problem is

$$xy = e^{-1-1/x}$$

$$y = e^{-(1+1/x)}/x.$$

Solve:

$$(e^y + 1)^2 e^{-y} dx + (e^x + 1)^3 e^{-x} dy = 0$$

First Order Ordinary Differential Equations:

$$\text{Solve } (e^{2y} - y) \cos x \frac{dy}{dx} = e^y \sin 2x, \quad y(0) = 0.$$

Dividing the equation by $e^y \cos x$ gives

$$\frac{e^{2y} - y}{e^y} dy = \frac{\sin 2x}{\cos x} dx.$$

Using identity $\sin 2x = 2 \sin x \cos x$, and integrating by parts

$$\int (e^y - ye^{-y}) dy = 2 \int \sin x dx$$

$$e^y + ye^{-y} + e^{-y} = -2 \cos x + c.$$

The initial condition $y = 0$ when $x = 0$ implies $c = 4$.

The solution is:

$$e^y + ye^{-y} + e^{-y} = 4 - 2 \cos x.$$

First Order Ordinary Differential Equations:

Reduction to Separable Form:

Certain non-separable ODEs can be made separable by transformations that introduce for y a new unknown function.

Suppose, $y' = f\left(\frac{y}{x}\right).$

Here, f is any (differentiable) function of y/x . The form of such an ODE suggests that we substitute

$$u=y/x$$

Thus, $y = ux$, and $y' = u'x + u.$

Substituting into the above ODE gives,

$$u'x + u = f(u) \text{ or } u'x = f(u) - u.$$

if $f(u) - u \neq 0$, this can be separated:

$$\frac{du}{f(u) - u} = \frac{dx}{x}.$$

Use method of
Sep. Vari., and
back
substitute

First Order Ordinary Differential Equations: Reduction to Separable Form:

$$2xyy' = y^2 - x^2.$$

Solution: We divide the given equation by $2xy$,

$$y' = \frac{y^2 - x^2}{2xy} = \frac{y}{2x} - \frac{x}{2y}.$$

Now substituting, $u=y/x$, or $y = ux$ and $y' = u'x + u$.

$$u'x + u = \frac{u}{2} - \frac{1}{2u},$$

$$u'x = -\frac{u}{2} - \frac{1}{2u} = \frac{-u^2 - 1}{2u}.$$

$$\frac{2u \, du}{1 + u^2} = -\frac{dx}{x}.$$

Separable

First Order Ordinary Differential Equations: Reduction to Separable Form:

Integrating,

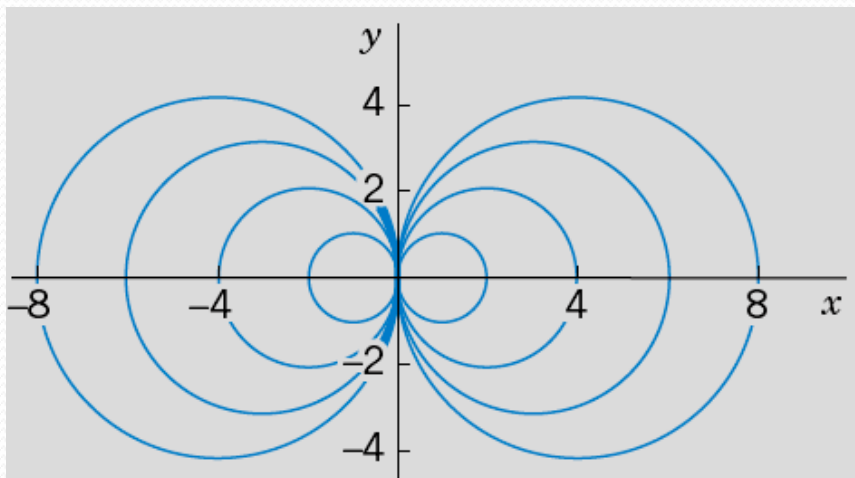
$$\ln(1 + u^2) = -\ln|x| + c^* = \ln\left|\frac{1}{x}\right| + c^*.$$

Taking exponents on both sides,

$$1 + u^2 = c/x$$

$$\text{or } 1 + (y/x)^2 = c/x.$$

Note: Multiply both sides of the solution by x^2 ,



$$x^2 + y^2 = cx.$$

Thus,

$$\left(x - \frac{c}{2}\right)^2 + y^2 = \frac{c^2}{4}.$$

Separable Equations: **Population Dynamics**

In Malthusian model , if $P(t)$ denotes the total population at time t , then this assumption can be expressed as

$$\frac{dP}{dt} \propto P \quad \text{or} \quad \frac{dP}{dt} = kP,$$

where k is a constant of proportionality.

$$\frac{1}{P} dP = k dt \text{ (separable variables)}$$

Integrating, $\ln|P| = kt + c$

$$P = P_0 e^{kt}, \quad \text{where } P_0 = e^c$$

Separable Equations: Population Dynamics

In a primary experiment, let the population of insects be N_0 . At time $t=4$ hours, the population is seen to be $2N_0$.

If the growth rate of the insects is proportional to the population of insects at certain time, how many hours later the population of the insects will be 8 times of the current population?

Solution: $N = N_0 e^{kt},$

Here, N_0 is the population at time $t=0$.

At time $t=4$ hours, the population is seen to be $2N_0$.

Hence, $2N_0 = N_0 e^{k \cdot 4}$

$$\text{or, } e^{4k} = 2 \quad \text{or, } 4k = \ln 2, k = \frac{\ln 2}{4} = 0.173$$

Hence, required time (T) for the population of the insects will be 8 times of the current population is,

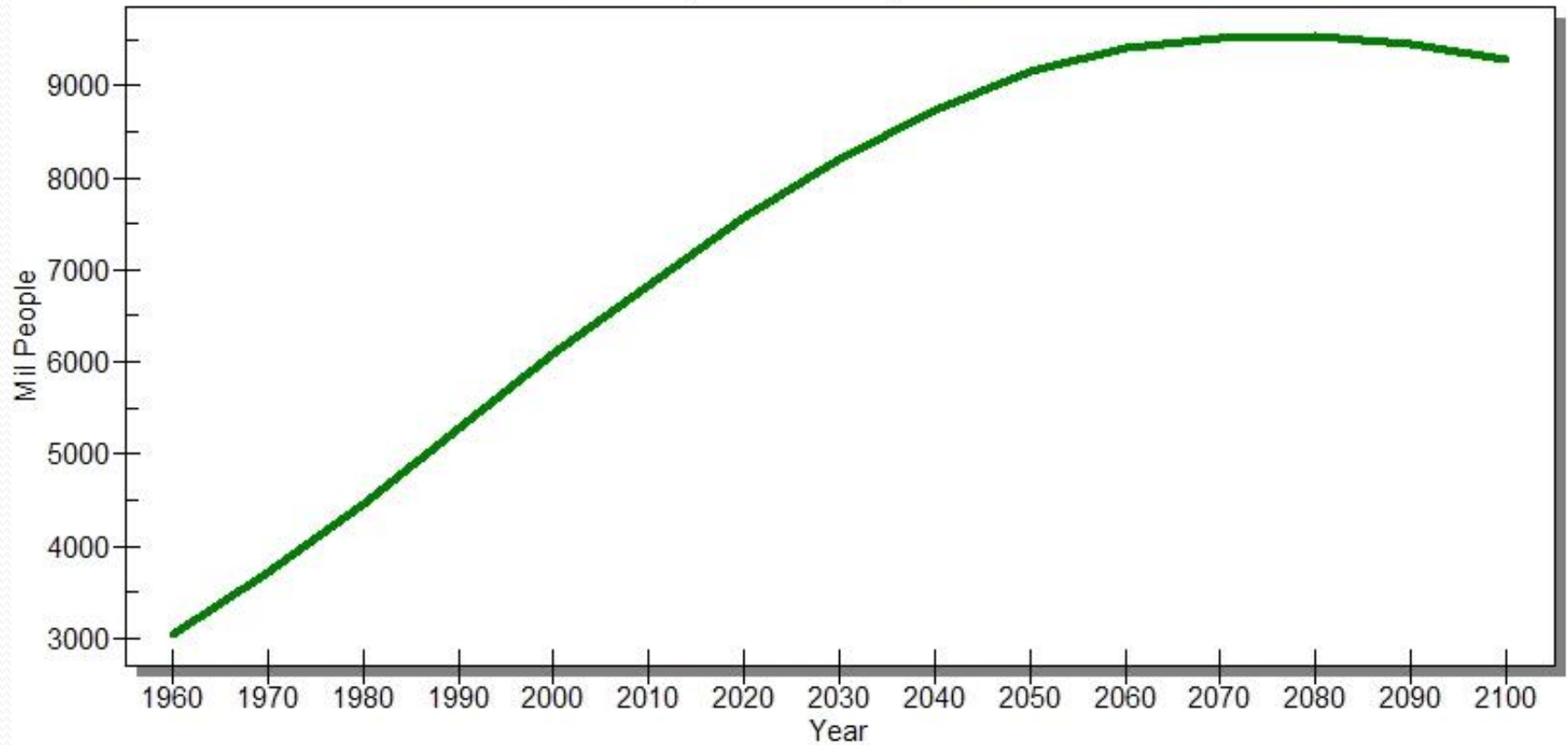
$$8N_0 = N_0 e^{0.173T}$$

$$e^{0.173T} = 8, T = \frac{\ln 8}{0.173} = 12.01 \approx 12H$$

Separable Equations: Population Dynamics

World Population

Actual to 2005, UN Medium Projection to 2100



Separable Equations: Heating Office Building (Newton's Law of Cooling)

Suppose that in winter the daytime temperature in a certain office building is maintained at 70°F. The heating is shut off at 10 P.M. and turned on again at 6 A.M.

On a certain day the temperature inside the building at 2 A.M. was found to be 65°F. Consider that the outside temperature varies between 50°F to 40°F by this time period. What was the temperature inside the building when the heat was turned on at 6 A.M.?

Solution: Let $T(t)$ be the temperature inside the building and T_m be the outside temperature (we consider here T_{out} is the average of 50°F to 40°F, that is $T_m = 45^\circ\text{F}$)

$$\frac{dT}{dt} \propto T - T_m \quad \text{or} \quad \frac{dT}{dt} = k(T - T_m),$$

$$\frac{dT}{T - 45} = k dt,$$

$$\ln |T - 45| = kt + c^*,$$

$$T(t) = 45 + ce^{kt} \quad (c = e^{c^*}).$$

General
solution

Separable Equations: Heating Office Building (Newton's Law of Cooling)

Particular solution: We choose 10 P.M. to be $t=0$. Then the given initial condition is $T(0)=70$ and yields a particular solution, call it T_p . By substitution

$$T(0) = 45 + ce^0 = 70,$$

$$c = 70 - 45 = 25,$$

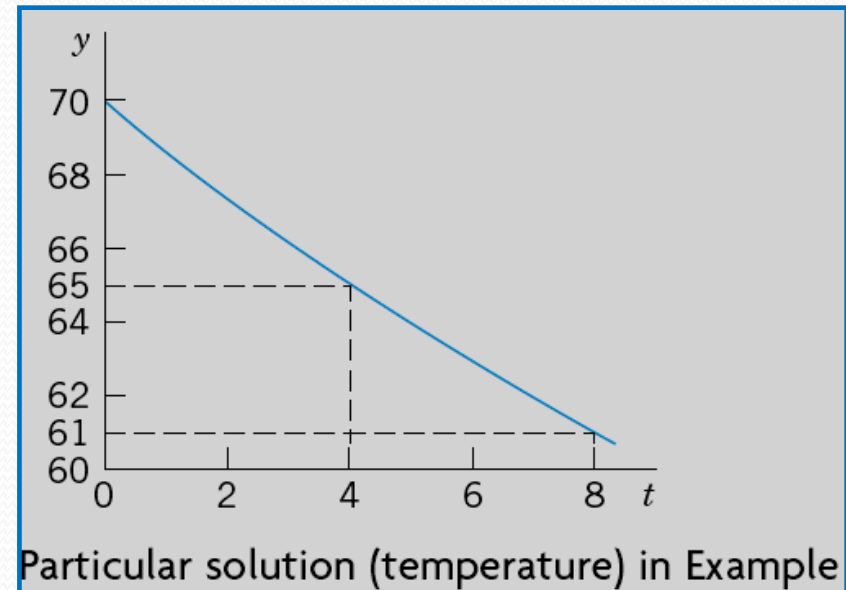
$$T_p(t) = 45 + 25e^{kt}.$$

We use $T(4)=65$, where $t=4$ is 2 A.M

$$T_p(4) = 45 + 25e^{4k} = 65,$$

$$e^{4k} = 0.8, \quad k = \frac{1}{4} \ln 0.8 = -0.056,$$

$$T_p(t) = 45 + 25e^{-0.056t}.$$



Particular
solution

First-order differential equation:

EXACT EQUATIONS

First-order differential equation: EXACT EQUATIONS

Definition: A first-order differential equation of the form

$$M(x, y) dx + N(x, y) dy = 0$$

is said to be an **exact equation** if the expression on the left-hand side is an exact differential.

A differential expression $M(x, y) dx + N(x, y) dy$ is an **exact differential** in a region R of the xy -plane if it corresponds to the differential of some function $f(x, y)$ defined in R .

Criterion for an Exact Differential:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

First-order differential equation: EXACT EQUATIONS

$$M(x, y) dx + N(x, y) dy = 0 \quad (1)$$

Solution Technique:

Consider $M(x, y) dx + N(x, y) dy = 0$, and $\frac{dM}{dy} = \frac{dN}{dx}$ holds true.
Let a function f be the solution of (1), such that

$$\frac{\partial f}{\partial x} = M(x, y). \quad (2)$$

We can find f by integrating $M(x, y)$ with respect to x while holding y constant:

$$f(x, y) = \int M(x, y) dx + g(y), \quad (3)$$

where the arbitrary function $g(y)$ is the “constant” of integration.

First-order differential equation: EXACT EQUATIONS

Now differentiate (3) with respect to y and assume that $\partial f / \partial y = N(x, y)$:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \int M(x, y) dx + g'(y) = N(x, y).$$

Hence,

$$g'(y) = N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx. \quad (4)$$

Finally, integrate (4) with respect to y and substitute the result in (3).
The implicit solution of the equation is $f(x, y) = c$.

Solve $2xy \, dx + (x^2 - 1) \, dy = 0$.

SOLUTION With $M(x, y) = 2xy$ and $N(x, y) = x^2 - 1$ we have

$$\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x}.$$

Thus the equation is exact.

The solution $f(x, y)$ is given by

$$f(x, y) = \int M(x, y) \, dx + g(y),$$

First-order differential equation: EXACT EQUATIONS

$$f(x, y) = x^2y + g(y).$$

The partial derivative of the above $f(x, y)$ with respect to y equal to $N(x, y)$.

$$\frac{\partial f}{\partial y} = x^2 + g'(y) = x^2 - 1. \quad \leftarrow N(x, y)$$

It follows that $g'(y) = -1$ and $g(y) = -y$.

Hence $f(x, y) = x^2y - y$

So, the solution is $x^2y - y = c.$

Note: The explicit form of the solution is $y = c/(1 - x^2)$ easily seen to be $y = c/(1 - x^2)$ and is defined on any interval not containing either $x = 1$ or $x = -1$.



Exercise 2.4 (Zill)

21. $(x + y)^2 dx + (2xy + x^2 - 1) dy = 0, \quad y(1) = 1$

Solution: Hints

Let $M = x^2 + 2xy + y^2$ and $N = 2xy + x^2 - 1$

$$M_y = 2(x + y) = N_x.$$

Therefore,

$$f = \frac{1}{3}x^3 + x^2y + xy^2 + h(y),$$

$$h'(y) = -1, \text{ and } h(y) = -y.$$

The solution is $\frac{1}{3}x^3 + x^2y + xy^2 - y = c$

If $y(1) = 1$ then $c = 4/3$

Hence, a solution of the initial-value problem is

$$\frac{1}{3}x^3 + x^2y + xy^2 - y = \frac{4}{3}.$$

First-order differential equation: EXACT EQUATIONS

Examples:

(1) $2xy \, dx + x^2 \, dy = 0$

(2) Solve $(e^{2y} - y \cos xy) \, dx + (2xe^{2y} - x \cos xy + 2y) \, dy = 0$.

(3) Solve $\frac{dy}{dx} = \frac{xy^2 - \cos x \sin x}{y(1 - x^2)}$, $y(0) = 2$.

(4) $\cos(x + y) \, dx + (3y^2 + 2y + \cos(x + y)) \, dy = 0$.

(5) $(x + y)^2 \, dx + (2xy + x^2 - 1) \, dy = 0$, $y(1) = 1$